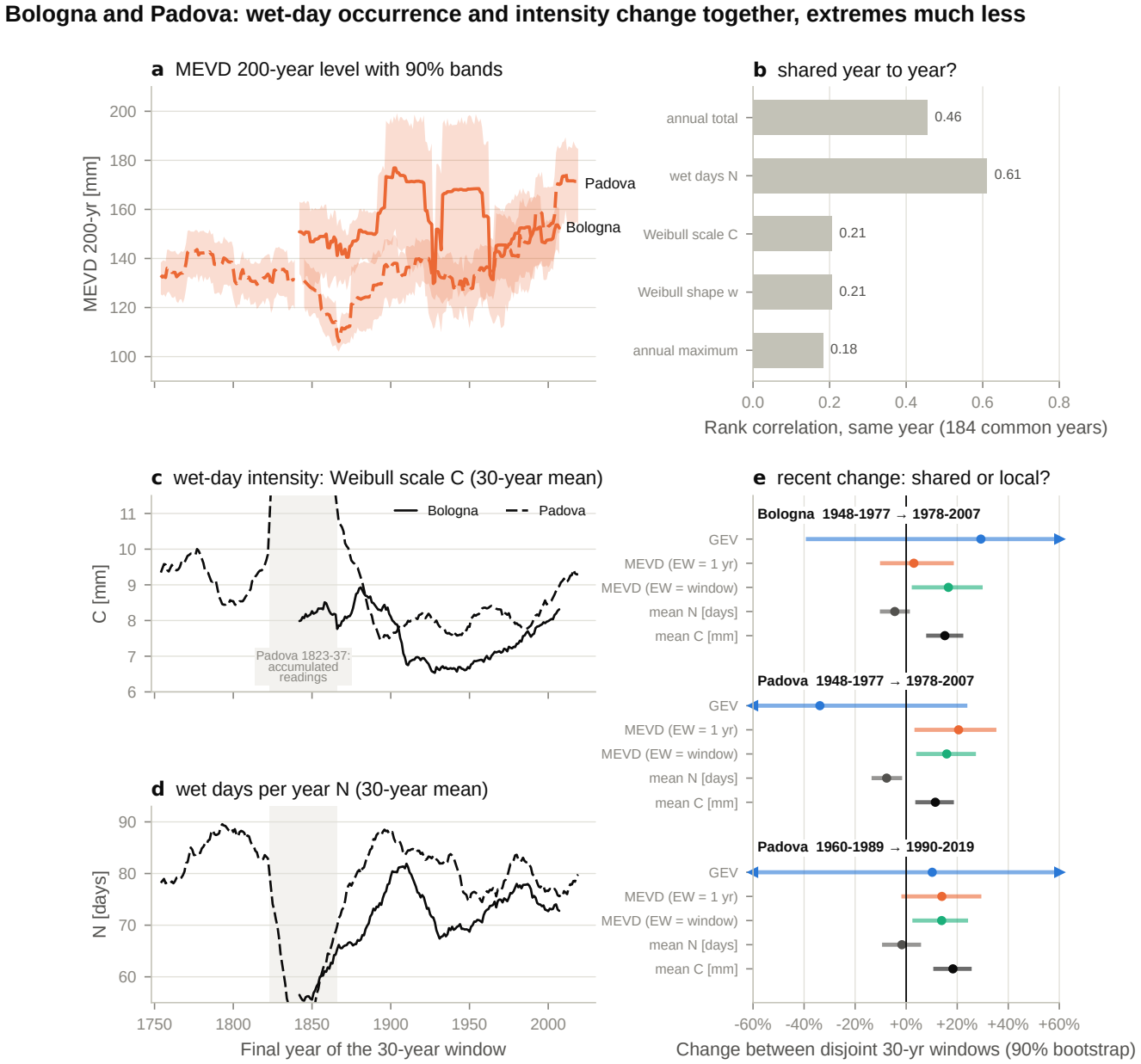
Bologna and Padova: wet-day occurrence and intensity change together, extremes much less
a MEVD 200-year level with 90% bands
MEVD 200-yr [mm]
200
180
160
140
120
100
Padova
Bologna
b shared year to year?
annual total
0.46
wet days N
0.61
Weibull scale C
0.21
Weibull shape w
0.21
annual maximum
0.18
0.0
0.2
0.4
0.6
0.8
Rank correlation, same year (184 common years)
c wet-day intensity: Weibull scale C (30-year mean)
Bologna
Padova
C [mm]
11
10
9
8
7
6
Padova 1823-37: accumulated readings
d wet days per year N (30-year mean)
N [days]
90
80
70
60
1750
1800
1850
1900
1950
2000
Final year of the 30-year window
e recent change: shared or local?
Bologna 1948-1977 → 1978-2007
GEV
MEVD (EW = 1 yr)
MEVD (EW = window)
mean N [days]
mean C [mm]
Padova 1948-1977 → 1978-2007
GEV
MEVD (EW = 1 yr)
MEVD (EW = window)
mean N [days]
mean C [mm]
Padova 1960-1989 → 1990-2019
GEV
MEVD (EW = 1 yr)
MEVD (EW = window)
mean N [days]
mean C [mm]
-60%
-40%
-20%
+0%
+20%
+40%
+60%
Change between disjoint 30-yr windows (90% bootstrap)